

MADHU PRAKASH S

Mechanical Designer

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🌐 linkedin.com/in/madhuprakashs 🔗 Portfolio LInk



Career Vision

Aspiring Mechanical Design Engineer with an M.Tech *in Mechanical Systems (Robotics & AI specialization)* and a B.E. in Mechanical Engineering. Hands-on experience in 3D CAD modelling, component design, and integrating custom-designed parts with standard industrial components to create functional assemblies. Familiar with mechanical design principles, manufacturing considerations, and basic application of GD&T and tolerance stack-up analysis. Knowledge of Six Sigma concepts at Green Belt level. Proficient in Autodesk Fusion 360 and actively developing skills in SolidWorks. Seeking an opportunity to contribute to a product design team while continuously building expertise in designing reliable and efficient engineering products.

Internship

Jan 2026 – Present
Bengaluru, India

Intern under CPED(Central Production and Engineering Department)

Wipro Hydraulics (Wipro Infrastructure Engineering)

- Studied the hydraulic cylinder sub-assembly line and analyzed 9 manual operations to identify automation opportunities, with direct involvement in improving 5 key activities.
- Proposed automation concepts to reduce manual effort, improve process consistency, and enhance assembly efficiency on the shop floor.
- Designed 3D CAD models for automation solutions including grippers, wiper blade mechanisms, and a trunnion tray fixture for sub-assembly applications.
- Prepared 2D manufacturing drawings for designed components with basic GD&T specifications, and utilized PowerPoint and Excel for project presentations and data tracking. Contributed to the shortlisting of 1 proposed concept for further consideration, validating the practicality of the design approach.
- **Impact:** Gained hands-on exposure to industrial manufacturing processes, mechanical design for automation, CAD detailing, and practical problem-solving in a real production environment.

Nov 2024 – Dec 2024
Bengaluru, India

Industrial Robotics Training

GTTC (Government Tool Room & Training Centre)

- Trained in robot operations, path planning, and trajectory tracing specifically for **KUKA Industrial Robots**.
- **Impact:** Gained hands-on experience in programming robotic systems for precision movement and industrial task execution.

Education

2023 – 2025
Bengaluru, India

Master of Technology in Robotics and Artificial Intelligence

Ramaiah Institute of Technology

8.97 cgpa

2019 – 2023
Bengaluru, India

Bachelor of Engineering in Mechanical

Dr. Ambedkar Institute Of Technology

8.0 cgpa.

Projects

Jan 2025 – Jul 2025

AI-Powered Handheld Thermal Imaging Device For Real-Time Human, Detection In Disaster Response Operations

PG Project

- Developed a helmet-mounted thermal camera system for real-time human detection in disaster response operations.

- Implemented detection using **Python, OpenCV, and YOLOv8 object detection** to identify victims through heat signatures, transmitting live data to remote command centers.
 - Designed the protective enclosure/mounting housing for the thermal camera and microcontroller using Autodesk Fusion 360 for secure helmet integration and field usability.
- Impact:** Secured **project funding** from the Alumni Association and successfully validated the system with performance results of **78% mean accuracy, 82% mean recall, and 78% mean precision**.

Nov 2022 – Jul 2023

Fabrication of Small Segment Car Operated by Electric Power

UG Major Project/VTU

- Led team of four members in end-to-end execution of foldable electric vehicle project, coordinating task allocation for design, fabrication, and final assembly.
- Designed vehicle body and chassis using AutoCAD and Autodesk Fusion 360; performed load calculations with Factor of Safety of 2 to establish maximum load capacity of 80 kg.
- Coordinated fabrication using 3 mm mild steel sheet and MS square tubes, supported welding and assembly, and conducted physical load validation testing to confirm structural integrity.
- Developed fully functional prototype with foldable form factor, demonstrating practical mobility and compact storage capability.

Impact: Successfully developed a fully functional prototype that could fold into a suitcase-like form factor, demonstrating practical mobility, compact storage, and project feasibility.

2026 – Present

Self-Initiated CAD Modelling Portfolio

Fusion 360 | 2026 – Present

- **Engineered a portfolio of 10+ multi-disciplinary 3D models**, ranging from heavy-duty industrial components to custom consumer products (desk organizers, display cases), RC automotive parts, and aerospace aircraft structures.
- Applied parametric modelling, assembly constraints, and part relationships across all models.
- Generated production-ready 2D technical drawings for assembly, ensuring accurate geometric specifications across various engineering domains.
- <https://maddy0330.github.io/portfolio> ↗ click here to view my portfolio.

Technical Skills

- **CAD & Design:** Autodesk Fusion 360, Catia V5, AutoCAD, SolidWorks, 3D part and assembly modelling, engineering drawings, GD&T, tolerance analysis, DFM/DFA
- **Manufacturing & Fabrication:** CNC programming (G-code/M-code), welding, mechanical assembly, load analysis and validation
- **Engineering & Process:** Mechanical design principles, manufacturing considerations, process optimization, problem-solving
- **Programming & Tools:** Python, VS Code, Google Colab
- **Quality & Operation:** Six Sigma and Lean Management

Certifications

Catia V5 Dassault Certification

Catia V5 Dassault Certification Preparation (In Progress, 2026)

Six Sigma and Lean Management

On-going training related to six sigma and lean management from CBEMG training institute

Declaration

I hereby declare that all information furnished above is authentic and accurate to the best of my belief.

Madhu Prakash S